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#include <iostream>
#include <cmath>
using namespace std;
double f(double x) {
  return x * x * x - 10 * x * x - 5 * x + 5;
}
double f_prime(double x) {
  return 3 * x * x - 20 * x - 5;
}
double newtonRaphson(double initial_guess, double tolerance, int max_iterations) {
  double x0 = initial_guess;
  double x1;
  for (int i = 0; i < max_iterations; i++) {
    double fx0 = f(x0);
    double fpx0 = f_prime(x0);
    if (fpx0 == 0) {
      cout << "Derivative is zero. No solution found." << endl;
      return NAN;
    }
        x1 = x0 - fx0 / fpx0;
    if (fabs(x1 - x0) < tolerance) {
      return x1;
    }
    x0 = x1;
  }
  return x1;
}
int main() {
  double initial_guess, tolerance;
  int max_iterations;
  cout << "Enter initial guess: ";
  cin >> initial_guess;
  cout << "Enter tolerance: ";
  cin >> tolerance;
  cout << "Enter maximum iterations: ";
  cin >> max_iterations;
  double root = newtonRaphson(initial_guess, tolerance, max_iterations);
  if (!isnan(root)) {
    cout << "The root is approximately: " << root << endl;
  }
  return 0;
}
```